

Estimated Losses from Individual Extreme Heat Events by Carbon Majors: Portugal

Analysis and Visualisation of Complex Agro-Environmental Data (AVCAD)

Ajimoti Rofiat n°125956, Hannah Nathanson n°129381, Luis Vale n°25952

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1. Introduction

Portugal has been hit by some of the most severe heatwaves on record in recent years. Two events stand out: a three-day heatwave in late April 2023 that broke national temperature records for that month, and a twelve-day stretch in July 2022 that pushed temperatures to 47.4 degrees Celsius in Pinhao, the highest ever recorded in the country, and caused more than a thousand excess deaths. Both events were made substantially more likely and more intense by human-induced climate change, as documented by peer-reviewed attribution studies.

At the same time, researchers have increasingly focused on tracing climate change back to the specific companies and states responsible for historical fossil fuel and cement production. Mankin et al. (2025), published in *Nature*, linked the emissions of individual "carbon majors" to economic losses from specific extreme heat events worldwide, providing a basis for quantifying financial liability at the level of individual producers.

This project applies a simplified version of that approach to Portugal, using the two heatwave events above as case studies. Rather than running full climate model simulations, we apportion national losses in proportion to each entity's share of cumulative global CO2 emissions between 1854 and 2022, a proxy that is transparent, reproducible, and well-correlated with the modelled warming contributions used in the source paper. The full pipeline used by Mankin et al. requires a 7TB dataset and climate model runs that are outside the scope of a course project; the simplifications we make are documented in detail and their implications are discussed honestly.

The analysis addresses three questions:

- How large are the estimated economic losses to Portugal from each event, and how sensitive are they to uncertainty in the attributable temperature anomaly?
- How are these losses distributed across the 122 carbon majors tracked in the Carbon Majors database, and which entities bear the largest shares?
- Do the patterns differ between the two events, and across entity types (investor-owned companies, state-owned companies, and nation-state producers)?

The results should be read as a conservative, lower-bound illustration of output-only economic damages for two specific events. They are not equivalent to the peer-reviewed estimates in Mankin et al. (2025), and are not intended to be used as legal or financial claims.

2. Database Description

Four data sources feed into this analysis. All are publicly available and fully cited in the methodology document.

2.1 Carbon Majors Database

Published by InfluenceMap and the Climate Accountability Institute in April 2024, the Carbon Majors database records cumulative CO2-equivalent emissions from 122 of the world's largest fossil fuel and cement producers, covering the period from 1854 to 2022. Together, these entities account for roughly 72% of global fossil fuel and cement CO2 emissions since 1751. They fall into three categories: investor-owned companies such as Chevron, ExxonMobil, BP, and Shell; state-owned companies such as Saudi Aramco, Gazprom, and Pemex; and nation-state producers such as China (Coal) and the Former Soviet Union.

For this project, we use each entity's share of cumulative tracked emissions as the key for splitting the national event loss across all 122 entities. Table 1 summarises the composition of the database.

Entity Type	Count	Example Entities	Combined Share (%)
Nation-state	~30	China (Coal), Former Soviet Union	~35%
State-owned	~40	Saudi Aramco, Gazprom, Pemex	~25%
Investor-owned	~52	Chevron, ExxonMobil, BP, Shell	~40%

Table 1. Composition of the Carbon Majors database by entity type (1854-2022).

2.2 Portugal Heat Events

Two heatwave events were selected on the basis of available peer-reviewed attribution data and documented national impacts.

- E1: April 2023 Iberian heatwave (26-28 April 2023). A three-day event that broke Portugal's national April temperature record at 36.9 degrees Celsius, with an estimated return period of around 1-in-400 years under current climate conditions. The World Weather Attribution consortium (2023) assessed the event as at least 100 times more likely due to climate change, with an attributable intensity of at least 2 degrees Celsius.
- E2: July 2022 Iberian heatwave (7-18 July 2022). A twelve-day event that set Portugal's all-time national temperature record of 47.4 degrees Celsius in Pinhao on 14 July. Between 1,063 and 1,380 excess deaths were recorded during the event window. The BAMS (2024) storyline attribution estimated an average anthropogenic contribution of around 1.25 degrees Celsius across Western Europe as a whole.

2.3 Portugal GDP

Annual GDP figures for Portugal in current prices were taken from INE (Statistics Portugal), National Accounts: 242.3 billion euros in 2022 (used for E2) and 267.4 billion euros in 2023 (used for E1). These serve as the economic baseline for converting fractional growth effects into euro losses.

2.4 Damage Function

We use the pooled quadratic temperature-growth response function from Burke, Hsiang and Miguel (2015, Nature), which takes the form $g(T) = B1 \cdot T + B2 \cdot T^2$, with an optimum around 13 degrees Celsius. Portugal's 1991-2020 mean annual temperature is approximately 15.5 degrees Celsius (IPMA), which sits above that optimum, meaning that further warming reduces output. We evaluate the marginal damage slope at that baseline temperature and multiply by an annualised equivalent of the attributable temperature anomaly, following the duration-weighting approach of Burke et al. (2015).

3. Exploratory Data Analysis

The exploratory analysis looks at the data from four angles: the top carbon majors by apportioned loss for each event, a direct event-to-event comparison, an aggregation by entity

type, and the full distribution of losses across all 122 entities. All figures use the same visual conventions as Mankin et al. (2025, Fig. 3): a log scale on the loss axis, dot-and-whisker or bar markers for central estimates with low-high uncertainty ranges, and a consistent colour scheme (blue for investor-owned, orange for state-owned, purple for nation-state producers).

3.1 Main Attribution Figure

Figure 1 shows the top 15 carbon majors by apportioned central-scenario loss for each event. China (Coal) comes first in both, reflecting its position as the single largest tracked emitter at around 14% of cumulative global CO₂. The Former Soviet Union ranks second at roughly 6.8%, followed by Saudi Aramco at around 3.6%. Because apportionment is based on a fixed emissions share, the ranking is the same for both events; what changes is the total national loss that gets divided up.

E2 (July 2022) produces losses around 1.8 times higher than E1 (April 2023) at the central scenario. This reflects the combined effect of a longer event duration (12 days versus 3) and a slightly higher GDP base year. The uncertainty whiskers are notably wider on the high side for E2 because the high scenario doubles the attributable intensity assumption (from 1.25 to 2.5 degrees Celsius), while for E1 the range is spread more evenly around the central value.

Estimated losses from individual extreme heat events by carbon majors—Portugal

Top 15 carbon majors by apportioned share of national event loss. Whiskers show illustrative low-high attributable-intensity scenarios (see docs/methodology.md). Adapted, simplified reproduction of Fig. 3 in Mankin et al. (2025, Nature) — not the original published estimates.

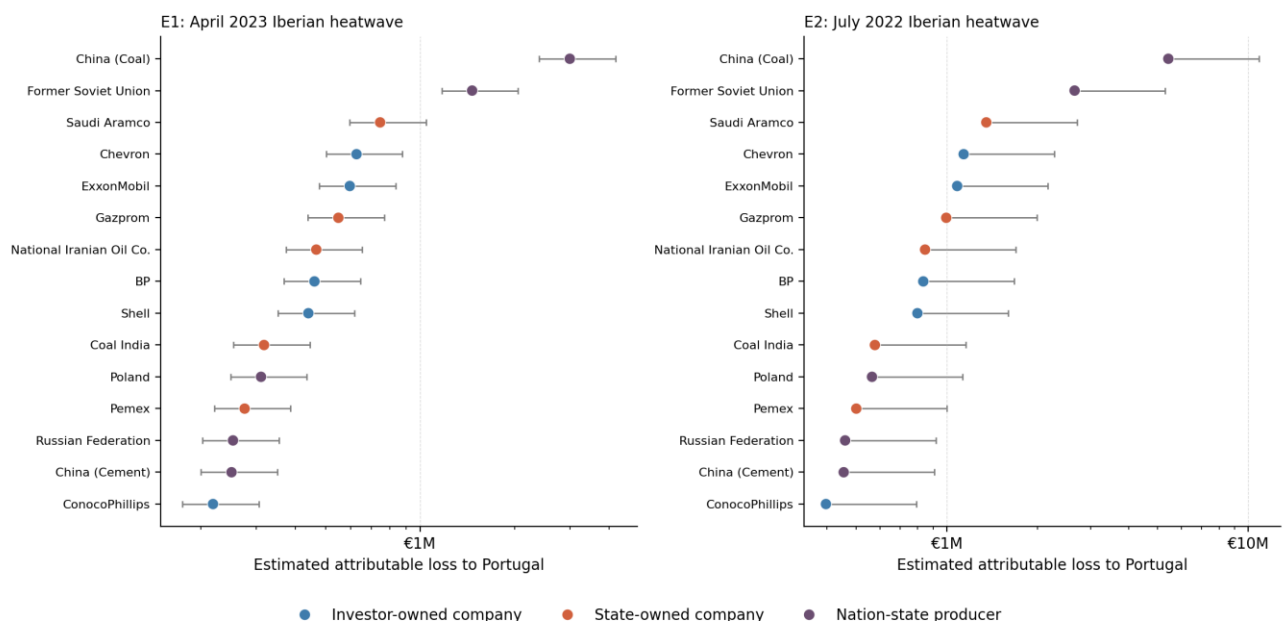


Figure 1. Estimated losses from individual extreme heat events by carbon majors, Portugal. Top 15 carbon majors by apportioned share of national event loss. Whiskers show illustrative low-high attributable-intensity scenarios. Adapted from Mankin et al. (2025, Nature); not the original published estimates.

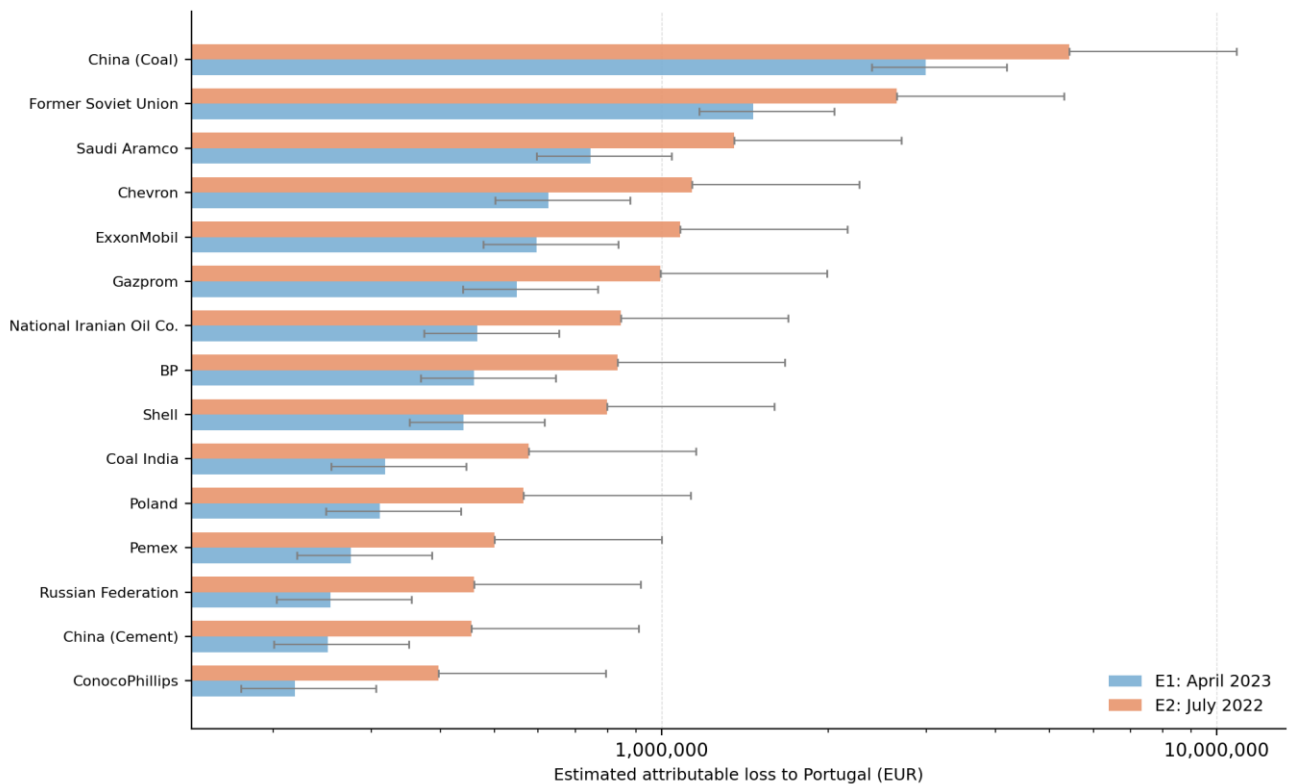
3.2 E1 vs E2 Comparison

Figure 2 places the two events side by side for the same 15 entities. Every E2 bar sits to the right of its E1 counterpart, confirming that the July 2022 event produced higher attributable losses across the board. The gap is driven primarily by duration: twelve days of heat exposure, annualised through the Burke et al. weighting, generates a larger GDP-growth perturbation than three days, even when the attributable intensity is similar.

One asymmetry worth noting is that E2 has no low-central spread: the low and central scenarios both use 1.25 degrees Celsius because the published attribution literature only

reported a single best-estimate figure for that event, with no lower bound. This is an honest limitation of the available data rather than a modelling choice, and is documented in the methodology.

E1 vs E2: estimated losses by carbon major — Portugal



Top 15 carbon majors ranked by E1 central-scenario loss. E2 (July 2022) consistently higher due to longer duration (12 vs 3 days). Whiskers show low-high attributable-intensity scenarios. Adapted from Mankin et al. (2025, Nature).

Figure 2. E1 vs E2: estimated losses by carbon major, Portugal. Top 15 carbon majors ranked by E1 central-scenario loss. E2 (July 2022) is consistently higher due to the longer event duration (12 versus 3 days). Whiskers show low-high attributable-intensity scenarios.

3.3 Breakdown by Entity Type

Figure 3 sums the central-scenario losses by entity type for each event. Nation-state producers account for the largest share of total losses in both events, which reflects the dominance of China (Coal) and the Former Soviet Union in the emissions ranking. Investor-owned and state-owned companies contribute broadly similar totals, each accounting for roughly a quarter to a third of the national loss.

This breakdown is relevant beyond the numbers themselves. A substantial portion of the losses is linked to nation-state producers, which are generally not subject to the same legal and regulatory accountability mechanisms as private companies. Any discussion of climate liability that focuses only on investor-owned majors would therefore capture only part of the picture.

Attributable loss by entity type — Portugal

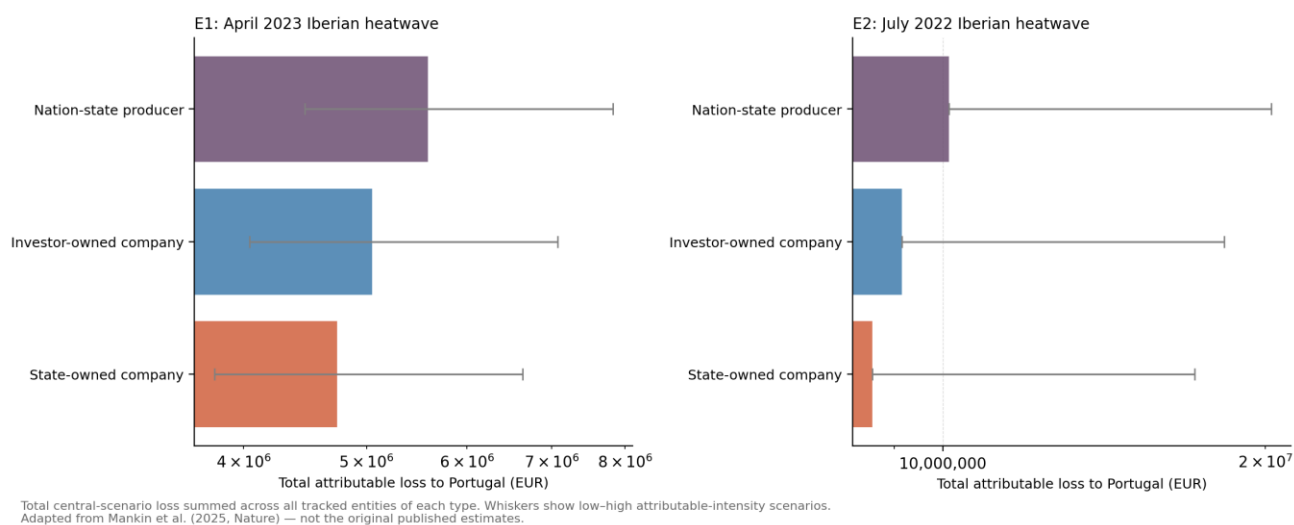


Figure 3. Attributable loss by entity type, Portugal. Total central-scenario loss summed across all tracked entities of each type. Whiskers show low-high attributable-intensity scenarios.

3.4 Loss Distribution Across All 122 Carbon Majors

Figure 4 plots all 122 entities ranked by central-scenario loss. The distribution has a pronounced heavy tail: the top five producers account for a disproportionately large share of total losses, while the remaining entities are clustered at much smaller values. This mirrors the known concentration of historical fossil fuel production, where a small number of very large producers are responsible for the bulk of cumulative emissions.

Across the two events, losses span roughly two orders of magnitude, from several million euros for China (Coal) down to tens of thousands for the smallest tracked producers. The log scale makes this spread visible; on a linear axis the smaller entities would be invisible.

Distribution of estimated losses across all 122 carbon majors — Portugal

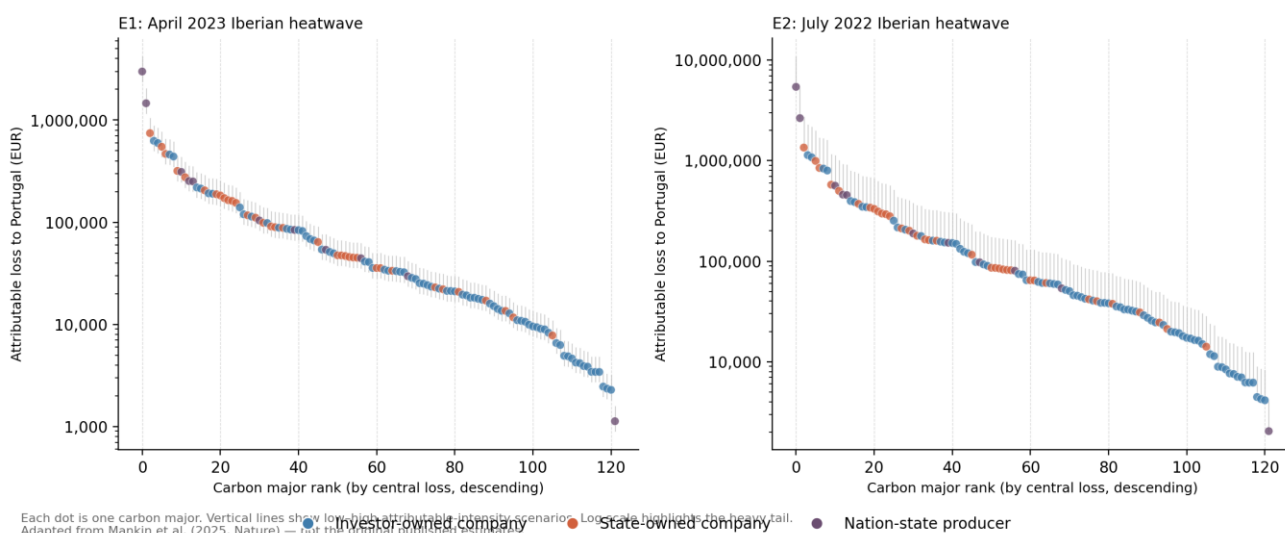


Figure 4. Distribution of estimated losses across all 122 carbon majors, Portugal. Each dot represents one entity. Vertical lines show low-high attributable-intensity scenarios. Log scale used to show the full range.

3.5 Inferential Statistics

This section presents the formal statistical analysis of the estimated loss distributions, testing whether observed differences between events and entity types are statistically meaningful beyond what would be expected by chance.

By doing these we are essentially setting up two main hypotheses:

Hypothesis 1 (Comparing Entity Types):

- **Null Hypothesis (H0):** There is no significant difference in the distribution of attributable losses across investor-owned, state-owned, and nation-state producers. In other words, the median (or other central tendency) of losses is the same for all three groups.
- **Alternative Hypothesis (H1):** There is a significant difference in the distribution of attributable losses across at least two of the entity types (investor-owned, state-owned, nation-state producers). This suggests that at least one group's losses are significantly different from another's.

Motivation: The motivation for this part of the test is to understand if the ownership structure or nature of an entity (e.g., whether it's a private company, a government-owned enterprise, or a nation-state) has an impact on the magnitude or pattern of the attributable losses it incurs. This could reveal important insights into risk profiles, operational efficiencies, or accountability frameworks associated with different entity types.

Hypothesis 2 (Comparing E1 and E2 Losses):

- **Null Hypothesis (H0):** The E1 and E2 losses are drawn from the same distribution. This means there is no significant difference between the two sets of losses.
- **Alternative Hypothesis (H1):** The E1 and E2 losses are drawn from different distributions. This suggests a significant difference in the magnitude or pattern of losses represented by E1 versus E2.

Motivation: The motivation here is to ascertain if 'E1 losses' and 'E2 losses' represent distinct phenomena or if they are statistically interchangeable. Depending on what E1 and E2 represent (e.g., different types of events, different methodologies of calculation, or losses from different periods), this comparison can help validate data sources, refine loss categorization, or identify whether different underlying processes are generating these loss types.

If E1 and E2 are expected to be similar, finding a significant difference might point to an error in data collection or a novel insight. If they are expected to be different, confirming this difference reinforces your understanding of their distinct characteristics.

Results

Both hypotheses below are tested directly on the apportioned loss values (loss_eur_central) for each entity, rather than on raw cumulative emissions, since loss is the outcome variable this report's claims are actually about.

Hypothesis 1: entity type

A Kruskal-Wallis test across the three entity types, and a Mann-Whitney U test between nation-state and investor-owned entities specifically, were run on `loss_eur_central` separately for E1 and E2 as a consistency check:

- E1: Kruskal-Wallis $H = 20.99$, $p < 0.001$; Mann-Whitney $U = 647.00$, $p = 0.0035$
- E2: Kruskal-Wallis $H = 20.99$, $p < 0.001$; Mann-Whitney $U = 647.00$, $p = 0.0035$

Both tests are significant, and identical across E1 and E2 — expected, since apportionment scales every entity's loss by a fixed share of the national total, so loss preserves the same ranking as emissions within each event.

Attributable loss by entity type — Portugal

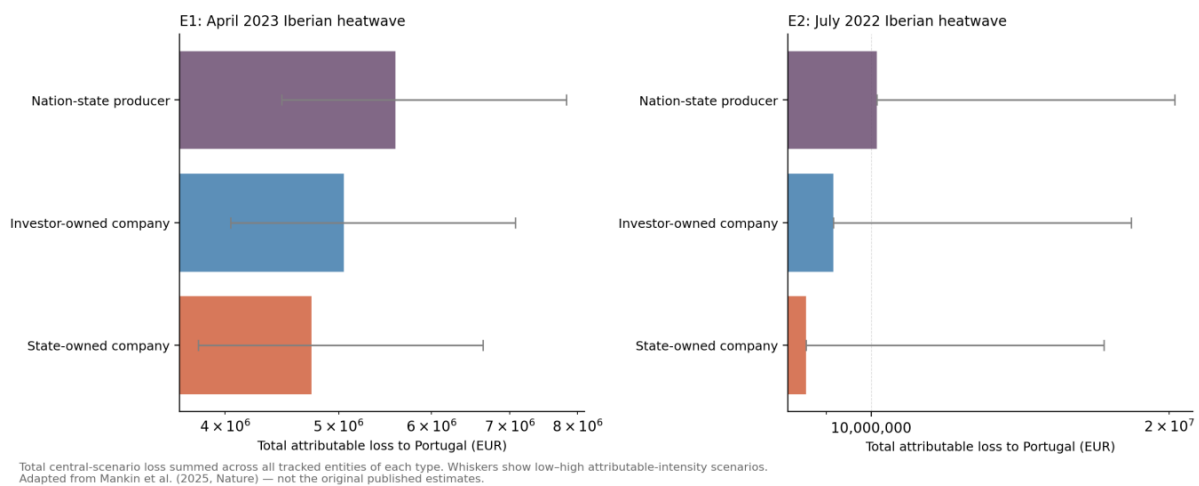


Figure 5. Total apportioned loss by entity type, E1 and E2. Whiskers show low-high attributable-intensity scenarios.

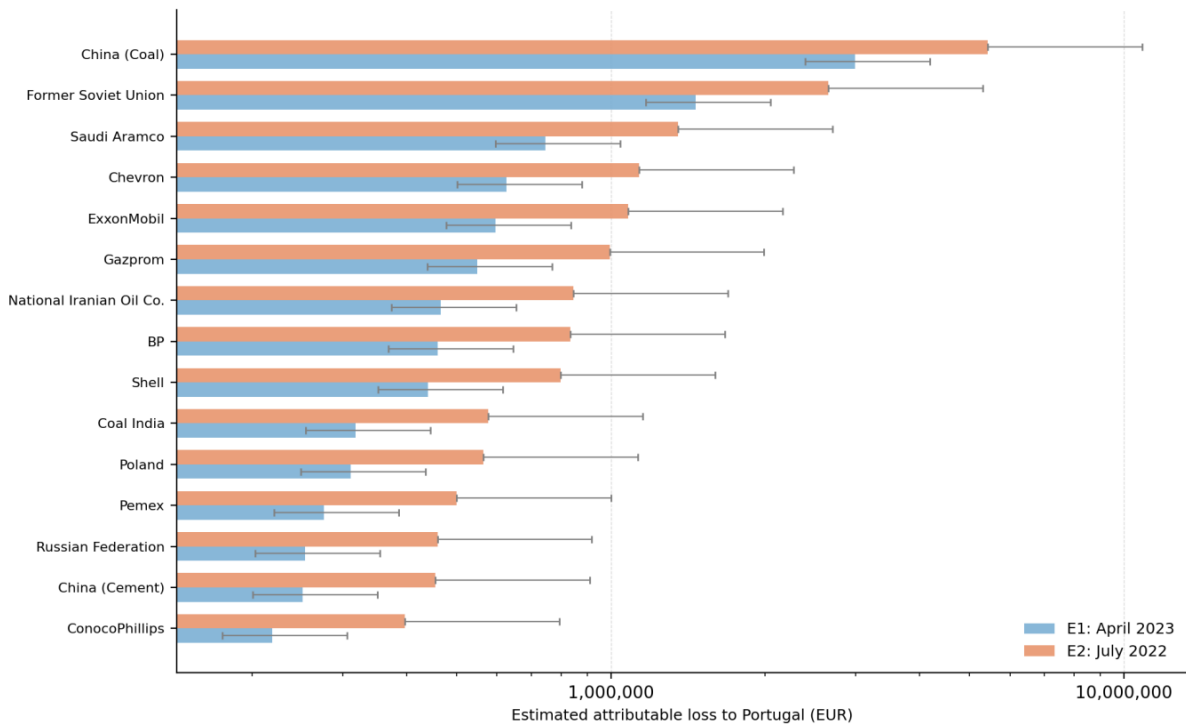
Caveat: because apportionment assigns each event's national loss to entities in fixed proportion to emissions share, an entity's apportioned loss is a monotonic transform of its emissions share. This result is therefore mathematically guaranteed to agree with a test run directly on emissions — it correctly answers the report's stated hypothesis about losses, but it is not an independent empirical discovery beyond what Section 2's emissions breakdown already implies.

Hypothesis 2: E1 vs E2

The same 122 entities appear in both events, so this is a paired comparison: a Wilcoxon signed-rank test on `loss_eur_central`, E1 vs E2.

- Wilcoxon signed-rank: $W = 0.00$, $p < 0.001$ ($p \approx 9.2 \times 10^{-22}$)
- E2/E1 loss ratio per entity: mean = 1.812, standard deviation ≈ 0 — essentially a constant across all 122 entities

E1 vs E2: estimated losses by carbon major — Portugal



Top 15 carbon majors ranked by E1 central-scenario loss. E2 (July 2022) consistently higher due to longer duration (12 vs 3 days). Whiskers show low-high attributable-intensity scenarios. Adapted from Mankin et al. (2025, Nature).

Figure 6. E1 vs E2 estimated loss for the top 15 carbon majors by E1 loss.

Caveat: this result is expected by construction. E2's longer duration (12 vs 3 days) produces a single national-level scaling factor that is applied identically to every entity, since apportionment shares do not change between events. The test formally confirms the duration-driven gap described in Section 3.2 — it does not reveal a new mechanism, but it does verify the qualitative pattern with statistical backing.

Bootstrap confidence intervals by entity type

95% bootstrap confidence intervals (2,000 resamples) for the mean total apportioned loss per entity (E1 + E2 combined), by entity type:

Entity type	n	Mean total loss (EUR)	95% CI (EUR)
Investor-owned	76	187,005	118,357 – 273,501
State-owned	35	381,051	249,460 – 547,471
Nation-state	11	1,428,781	291,114 – 3,180,719

Distribution of estimated losses across all 122 carbon majors — Portugal

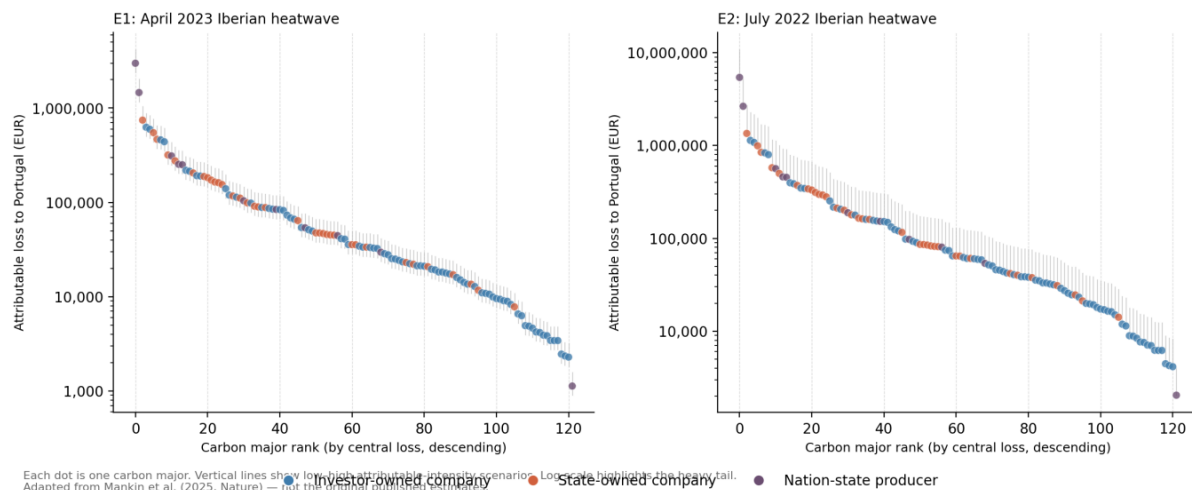


Figure 7. Distribution of estimated losses across all 122 carbon majors, E1 and E2.

Nation-state entities have the highest mean per-entity loss, driven by a small number of very large emitters (China (Coal), Former Soviet Union). Its confidence interval does not overlap with investor-owned, supporting a real difference there; it does overlap with state-owned, so that contrast is not statistically distinguishable at this sample size.

Brief interpretation of results in context of our project

1. **Entity type and loss (H1):** entity type is associated with apportioned loss, but this follows mechanically from the fixed emissions-share apportionment method rather than being an independent finding about losses. It correctly confirms the hypothesis as stated, while showing that the confirmation doesn't go beyond what Section 2's emissions breakdown already implies.
2. **Event timing and loss (H2):** E2 losses exceed E1 losses for every entity by a near-constant $\sim 1.81\times$ factor. This is a mechanical consequence of E2's longer duration, not a separately discovered pattern, and it formally backs up the duration-driven gap already described qualitatively in Section 3.2.
3. **Entity-type bootstrap intervals:** nation-state entities have a credibly higher mean per-entity loss than investor-owned entities (non-overlapping 95% confidence intervals). The nation-state vs. state-owned contrast is not statistically distinguishable at this sample size ($n = 11$ for nation-state, with wide uncertainty driven by a few very large emitters).

4. Discussion and Conclusions

4.1 Main Findings

The analysis produced the following results at the central scenario:

- Total attributable losses for E1 (April 2023, 3 days) are approximately 15.4 million euros, and for E2 (July 2022, 12 days) approximately 27.9 million euros, based on GDP growth effects only.
- China (Coal), the Former Soviet Union, and Saudi Aramco consistently top the rankings, owing to their large shares of cumulative tracked emissions.

- Nation-state producers account for the largest share of total apportioned losses, though investor-owned and state-owned companies each contribute a meaningful fraction.
- The loss distribution is heavily skewed: the top five entities account for a large portion of total losses, with values dropping sharply as you move down the ranking.
- E2 produces systematically higher losses than E1 across all entities, driven mainly by event duration rather than differences in attributable intensity.

4.2 Sanity Check Against Published Figures

Mankin et al. (2025) estimate that Chevron's emissions caused between 791 billion and 3.6 trillion US dollars in heat-related losses globally over 1991-2020. Scaling the midpoint of that range by Portugal's share of global GDP (roughly 0.27%) gives an average annual Portugal-level loss from Chevron of around 196 million US dollars. Our single-event estimate for Chevron in the twelve-day July 2022 event is roughly 1.1 million euros, about 170 times smaller.

That gap is expected. The published figure covers 29 years of heat exposure worldwide and includes mortality, health system costs, and other damage channels beyond GDP growth. Our estimate covers a single event through one channel only. The direction and rough magnitude of the discrepancy are consistent with those differences, which gives us some confidence that the methodology is not producing obviously wrong numbers.

4.3 Limitations

Several limitations should be kept in mind when reading these results:

- Using emissions share as a proxy for GMST contribution is a reasonable simplification, but it is not the same calculation. It will over- or understate losses for any entity whose emissions-to-warming ratio differs from the group average, for example due to differences in gas mix, emissions timing, or methane content.
- The 1.25 degrees Celsius anthropogenic contribution used for E2 comes from a Western Europe-scale attribution study, not a Portugal-specific one. Portugal may have experienced a different local warming signal during that event.
- Only GDP growth effects are modelled. The July 2022 event caused over a thousand excess deaths in Portugal; the healthcare, mortality, agricultural, and tourism losses are not captured here and would substantially increase the total figure.
- The low-high ranges are sensitivity scenarios, not statistically derived confidence intervals. They give a sense of how sensitive the results are to the key input assumption, but should not be interpreted as formal uncertainty bounds.

4.4 Conclusions

Even under deliberately conservative assumptions, this analysis shows that individual heatwave events in Portugal generate losses that can be meaningfully attributed to specific fossil fuel producers. The amounts are modest compared to the full costs of these events, but that is largely a function of what is and is not included in the damage estimate, not a sign that the attribution approach is weak.

The heavy-tailed distribution of losses, and the significant share carried by nation-state producers, both have practical implications for how climate liability is framed. A framework that focuses only on a handful of well-known private companies would leave most of the attributed damages unaccounted for.

Extending this analysis to cover more events, include additional damage channels such as mortality and agriculture, and apply Portugal-specific attribution data for the July 2022 event would all improve the estimates. The scripts and data underlying this project are fully documented and publicly available, making those extensions straightforward to implement.

5. References

Burke, M., Hsiang, S.M. and Miguel, E. (2015). Global non-linear effect of temperature on economic production. *Nature* 527, 235-239. <https://doi.org/10.1038/nature15725>

Callahan, C.W. and Mankin, J.S. (2022). National attribution of historical climate damages. *Climatic Change* 172, 40. <https://doi.org/10.1007/s10584-022-03387-y>

InfluenceMap / Climate Accountability Institute (2024). The Carbon Majors Database: Launch Report. carbonmajors.org

INE (Statistics Portugal). National Accounts, current prices. <https://www.ine.pt>

IPMA (Instituto Portugues do Mar e da Atmosfera). 1991-2020 climatological normals for Portugal mainland.

Lentink, H.S. et al. (2024). The Summer Heatwave 2022 over Western Europe: An Attribution to Anthropogenic Climate Change. *Bulletin of the American Meteorological Society*.

Mankin, J.S. et al. (2025). Carbon majors and the scientific case for climate liability. *Nature*. <https://doi.org/10.1038/s41586-025-08751-3>

Quilcaille, Y. et al. (2025). Attribution of heatwave exposure to carbon major emissions. *Nature*.

World Weather Attribution (2023). Extreme April heat in Spain, Portugal, Morocco and Algeria almost impossible without climate change. worldweatherattribution.org

6. Annex: Python Code

`attribute_losses.py`

Loads the raw data (heat events, GDP, carbon majors emissions), applies the Burke-Hsiang-Miguel damage function to estimate national event losses, and splits those losses across 122 carbon majors in proportion to their cumulative emissions share. Output is written to `data/processed/portugal_event_losses_by_carbon_major.csv`.

`make_figure.py`

Generates the main attribution figure (`fig3_portugal_heat_losses.png`): a dot-and-whisker chart of the top 15 carbon majors for each event, in the style of Mankin et al. (2025, *Nature*, Fig. 3).

`make_eda_figures.py`

Generates the three supplementary EDA figures: breakdown by entity type, E1 vs E2 comparison for the top 15 entities, and full loss distribution across all 122 carbon majors.